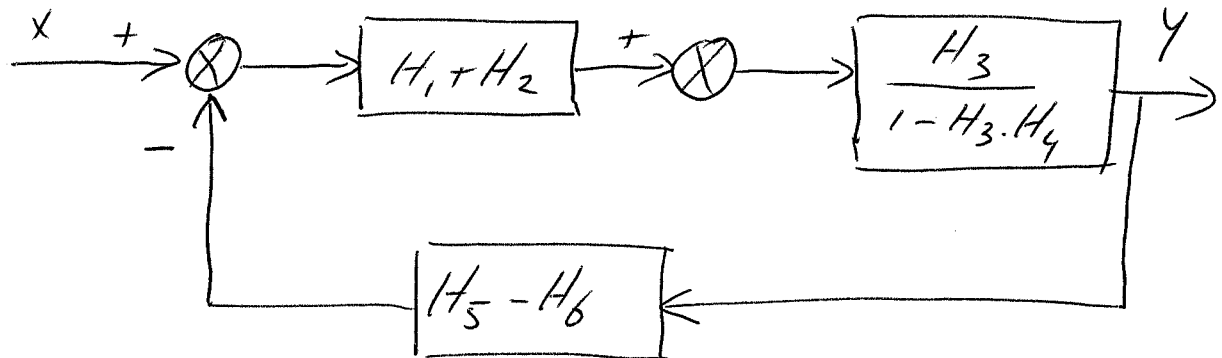


§ 2.8 Opdraven

1a) gebruik: $H = \frac{\text{recht doorgaande}}{1 + \text{rondgaande}}$
antwoorden: zie achter in het boek

b.) eerst vereenvoudigen



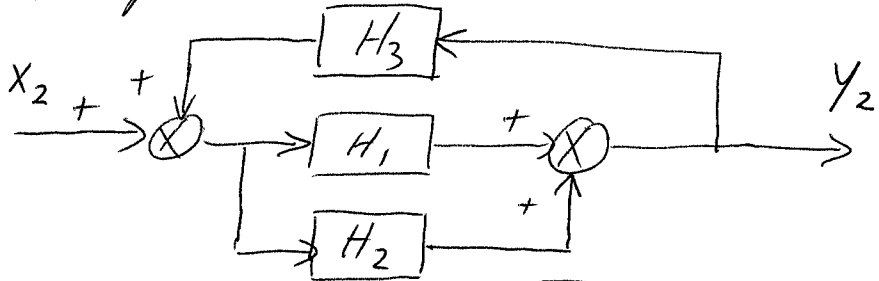
dan: $H = \frac{\text{recht doorgaande}}{1 + \text{rondgaande}}$

$$H = \frac{(H_1 + H_2) \cdot \frac{H_3}{(1 - H_3 \cdot H_4)}}{1 + \frac{(H_1 + H_2) \cdot H_3}{(1 - H_3 \cdot H_4)} \cdot (H_5 - H_6)}$$
$$= \frac{(H_1 + H_2) \cdot H_3}{1 - H_3 \cdot H_4 + (H_1 + H_2) \cdot H_3 \cdot (H_5 - H_6)}$$

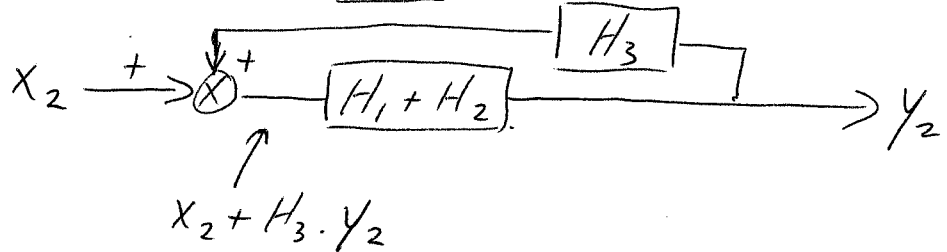
§ 2.8 opgaven

1c. eerst vereenvoudigen

dit deel:



wordt:



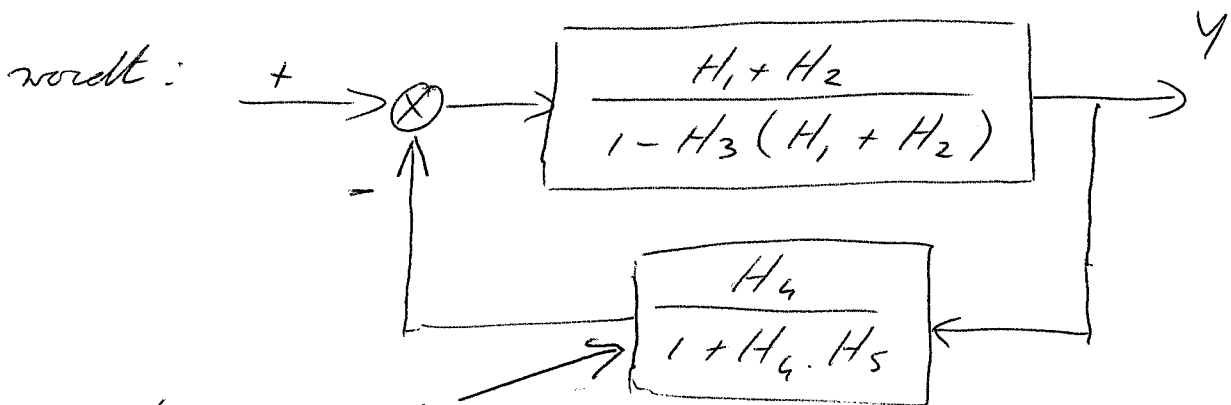
$$(x_2 + H_3 \cdot y_2)(H_1 + H_2) = y_2$$

$$x_2(H_1 + H_2) + y_2 \cdot H_3 \cdot (H_1 + H_2) = y_2$$

$$y_2 - y_2 \cdot H_3(H_1 + H_2) = x_2(H_1 + H_2)$$

$$y_2(1 - H_3(H_1 + H_2)) = x_2(H_1 + H_2)$$

$$\frac{y_2}{x_2} = \frac{H_1 + H_2}{1 - H_3(H_1 + H_2)}$$



terugkoppeling, dus:

$$H = \frac{\text{recht door}}{1 + \text{rondgaand}}$$

§ 2.8 opgaven

vervolg 1 c

$$\begin{aligned} H_{\text{totaal}} &= \frac{\text{recht door}}{1 + \text{rondgaand}} \\ &= \frac{(H_1 + H_2)}{1 - H_3 \cdot (H_1 + H_2)} \\ &= \frac{1 + (H_1 + H_2)}{1 - H_3 \cdot (H_1 + H_2)} \cdot \frac{H_4}{1 + H_4 \cdot H_5} \\ &= \frac{(H_1 + H_2)(1 + H_4 \cdot H_5)}{(1 + H_4 \cdot H_5) \cdot (1 - H_3 \cdot (H_1 + H_2)) + (H_1 + H_2) \cdot H_4} \end{aligned}$$